

COMPLETE GUIDE

The 50 RAG Terms Behind Great Systems

Essential concepts for building and evaluating
retrieval-augmented generation systems

RAG

Retrieval

Embeddings

Vector Store

Evaluation



Naved Khan
Senior Gen-AI Engineer

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1. RAG

Retrieval-Augmented Generation. It augments LLM answers with retrieved external documents.

2. Retrieval

Fetching the most relevant documents or chunks from a store in response to a user query.

3. Embeddings

Dense vector representations of text that capture semantic meaning for similarity search.

4. Vector Store

Database optimized for storing and querying high-dimensional embedding vectors.

5. Chunking

Splitting documents into smaller segments for indexing and retrieval in RAG.



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6. Context Window

Maximum input length in tokens the LLM can use for prompt plus retrieved context.

7. Semantic Search

Finding documents by meaning rather than exact keyword match.

8. Hallucination

LLM producing plausible but false or unsupported information.

9. Dense Retrieval

Retrieval using embeddings and vector similarity such as cosine similarity.

10. Sparse Retrieval

Keyword-based retrieval using term frequency and matching, for example BM25.



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11. Hybrid Search

Combining dense vector and sparse keyword retrieval for better recall and precision.

12. Reranking

Re-scoring top retrieved results with a cross-encoder or similar model for better order.

13. BM25

Sparse retrieval algorithm using term frequency and inverse document frequency.

14. Query

User question or request used to retrieve and generate an answer.

15. Document Loader

Component that loads raw documents from files, APIs, or DBs into a standard format.



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16. Text Splitter

Logic that splits documents into chunks by size, sentence, or semantics.

17. Embedding Model

Model that converts text into fixed-size vectors, for example sentence-transformers.

18. Indexing

Process of chunking, embedding, and storing chunks in a vector or search index.

19. Similarity Search

Finding vectors closest to the query vector, for example k-NN or approximate nearest neighbor.

20. Cosine Similarity

Common metric for comparing embedding vectors in retrieval.



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21. Context Compression

Reducing size of retrieved context while keeping key information, for example summarization.

22. Query Expansion

Generating multiple or refined queries from the original to improve retrieval.

23. Query Transformation

Rewriting or decomposing the user query before retrieval, for example for multi-hop.

24. Naive RAG

Basic RAG with a single retrieval step then generation, no reranking or query refinement.

25. Advanced RAG

RAG with query refinement, reranking, or multiple retrieval steps.



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26. Self-RAG

RAG where the model decides when to retrieve and reflects on retrieval and generation.

27. Corrective RAG

RAG that validates retrieval and may re-retrieve or regenerate when needed.

28. Parent Document Retriever

Retrieve small chunks for matching but return larger parent chunks for context.

29. Chunk Overlap

Overlapping tokens between adjacent chunks to preserve context at boundaries.

30. MMR (Maximal Marginal Relevance)

Balancing relevance and diversity when selecting retrieved passages.



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31. Cross-Encoder

Model that scores query-document pairs for reranking; slower but more accurate.

32. Bi-Encoder

Encoder that produces separate query and document embeddings for fast retrieval.

33. HyDE (Hypothetical Document Embeddings)

Embedding a model-generated hypothetical answer to improve retrieval.

34. Multi-Query Retrieval

Running several query variants and merging or deduplicating results.

35. k-NN

k-nearest neighbors: retrieving the k vectors closest to the query vector.



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36. HNSW

Approximate nearest neighbor index used in many vector stores.

37. Faithfulness

Metric: generated answer is consistent with retrieved context, no unsupported claims.

38. Relevance

Metric: retrieved documents or generated answer match the query intent.

39. Grounding / Citation

Attributing generated claims to specific retrieved passages or sources.

40. Retrieval Quality

How well retrieval satisfies the information need in relevance and coverage.



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41. Context Relevance

How relevant the retrieved context is to the query.

42. Answer Relevance

How well the generated answer addresses the query.

43. MRR (Mean Reciprocal Rank)

IR metric: reciprocal of the rank of the first relevant document.

44. NDCG

Normalized Discounted Cumulative Gain; measures ranking quality.

45. Token Limit

Maximum tokens the model can accept; constrains context and generation.



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46. RAG Pipeline

End-to-end flow: load, chunk, embed, index, retrieve, optional rerank, then generate.

47. Production RAG

RAG built for scale, latency, monitoring, and reliability in production.

48. Document / Chunk

Unit of text such as a passage or section stored and retrieved in RAG.

49. System Prompt

Instructions that define the RAG assistant role and how to use retrieved context.

50. Generation

LLM producing the final answer using the prompt and retrieved context.



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